

SECTION 2 – ROOM LAYOUTS

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2-1 INTRODUCTION

When laying out a floor plan there are special considerations that must be taken into account due to the magnetic field effect on equipment. The maximum magnetic field in which the upgrade supplied equipment can be located is listed in Table 2-1.

TABLE 2-1
PROXIMITY LIMITS

GAUSS (mT) LIMIT (See Note 1)	EQUIPMENT
50 GAUSS (5mT) OR LESS	Water Chiller for Gradient Coil Cooling
Note 1 Recommended limits given above are based on general MR site planning guidelines. Actual susceptibility of specific devices may vary significantly depending on electrical design orientation of the device relative to the magnetic field and the degree of interference considered unacceptable.	

2-2 MINIMUM DOOR/HALLWAY SIZES

Table 2-2 lists minimum actual clearance opening dimensions for doors and hallways required by new equipment installed as part of the Upgrade. Installation or replacement of components listed in Table 2-3 must be taken into consideration when determining delivery route hallway and door dimensions. Clearance for maneuvering around corners or turns must also be taken into consideration. Refer to SECTION 8, SHIPPING AND DELIVERY DATA, for component shipping dimensions.

TABLE 2-2
MINIMUM HALLWAY/DOOR DIMENSIONS

COMPONENT	MINIMUM HALLWAY/ DOOR WIDTH* in. (mm)	MINIMUM HALLWAY/ DOOR HEIGHT* in. (mm)	COMMENTS
Front/Rear Arc Covers	36 (914)	78 (1981)	See illustrations in Section 2-6 COMPONENT DIMENSIONS.
Front/Rear Endbells	36 (914)	78 (1981)	See illustrations in Section 2-6 COMPONENT DIMENSIONS.
Note * Minimum hallway and door dimensions are actual clearance openings. Width and height of rigging equipment is not included in above dimension.			

TABLE 2-3
UPGRADE COMPONENT DIMENSIONS FOR INSTALLATION/REPLACEMENT

COMPONENT	APPROXIMATE WEIGHT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	COMMENTS
RF Coil in shipping case	275 (125)	38 x 38 x 77 (965 x 965 x 1956)	Weight and dimension include coil, case, and shipping pallet. RF Coil Case has wheels and handles for ease moving after removal from pallet. Dimensions of RF Coil Case without pallet are 34 x 34 x 73 (864 x 864 x 1854).

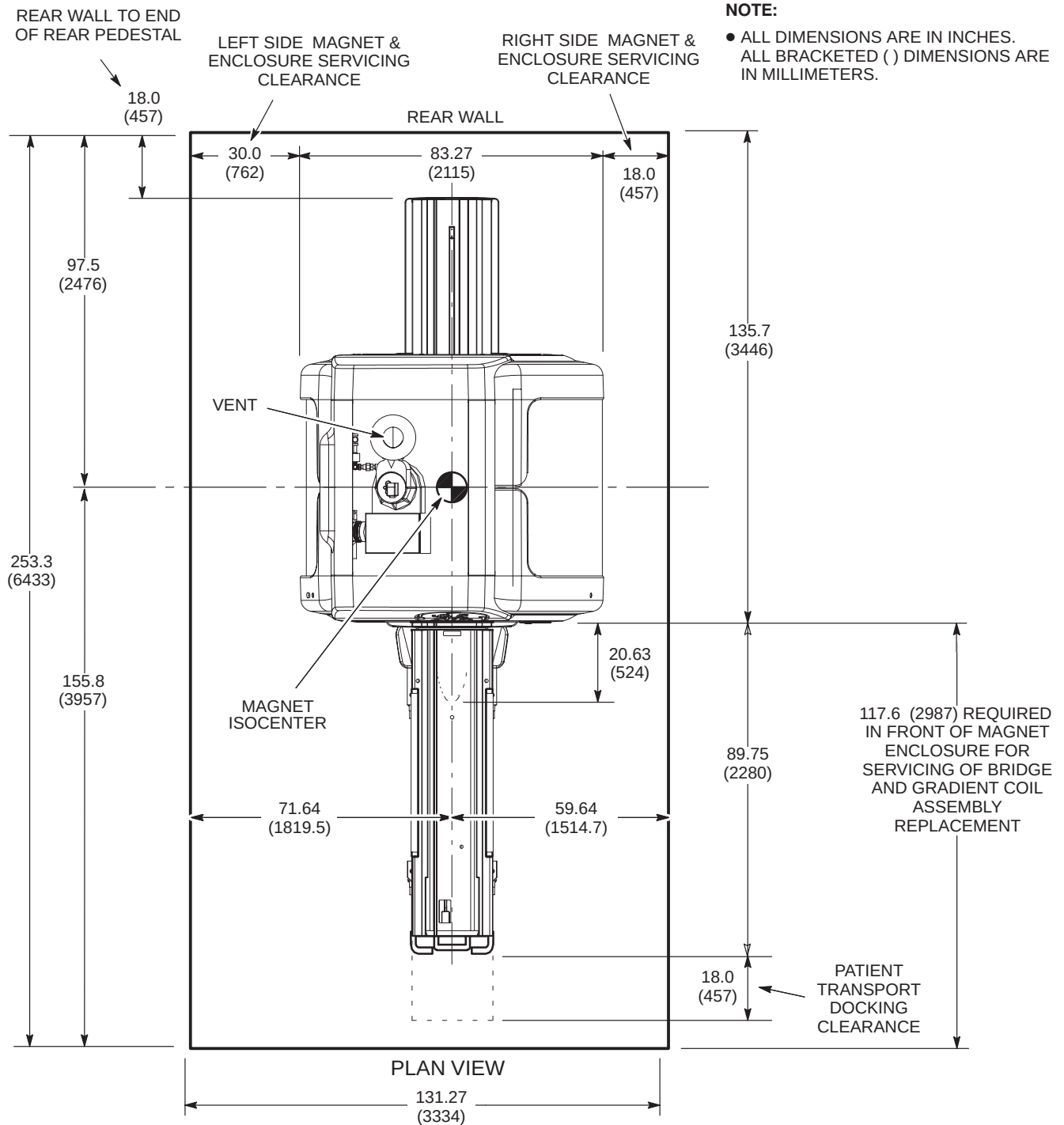
2-3 CABLING CONSIDERATIONS

Cable runs in the Magnet Room as well as throughout the system must be in accordance with local and national codes. Cabinets cable access areas are shown on illustrations in Section 2-6, COMPONENT DIMENSIONS. Some cable routing will be changed due to new equipment depending on system configuration being upgraded. Refer Section 6, INTERCONNECT DATA, for cables and their dimensions information.

Note

The WideOpen Enclosure has different Rear Pedestal cable access dimensions compared to previous configuraiton. See Illustration 2-3 for Magnet cable access areas information.

2-3 CABLING CONSIDERATIONS (Continued)



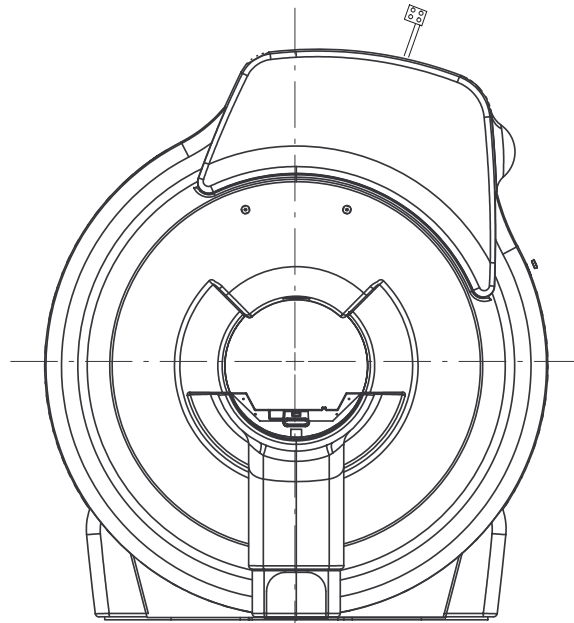
1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET WITH WIDEOPEN ENCLOSURE (MINIMUM SERVICE AREA)

ILLUSTRATION 2-1

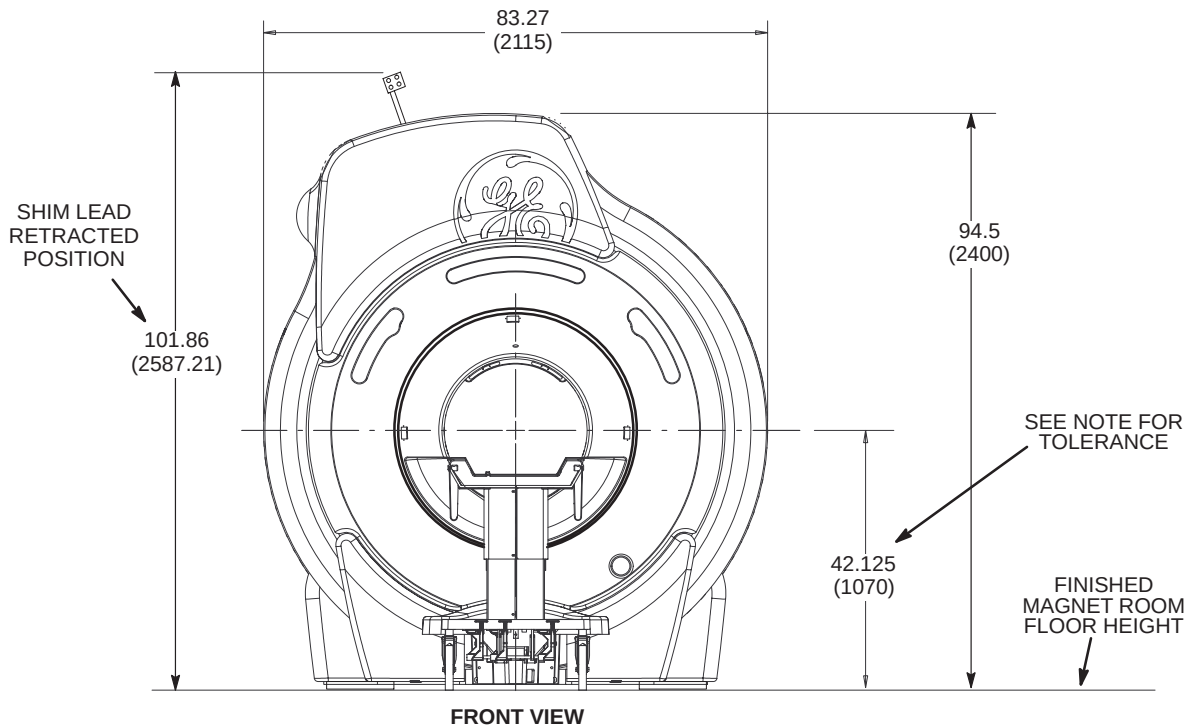
2-3 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- FINISHED FLOOR TO MAGNET CENTER LINE HEIGHT DIMENSION MUST BE 42.125 ± 0.25 in. (1070 ± 6.35 mm) TO ALLOW ENCLOSURE COVERS TO MOVE FREELY.



REAR VIEW



FRONT VIEW

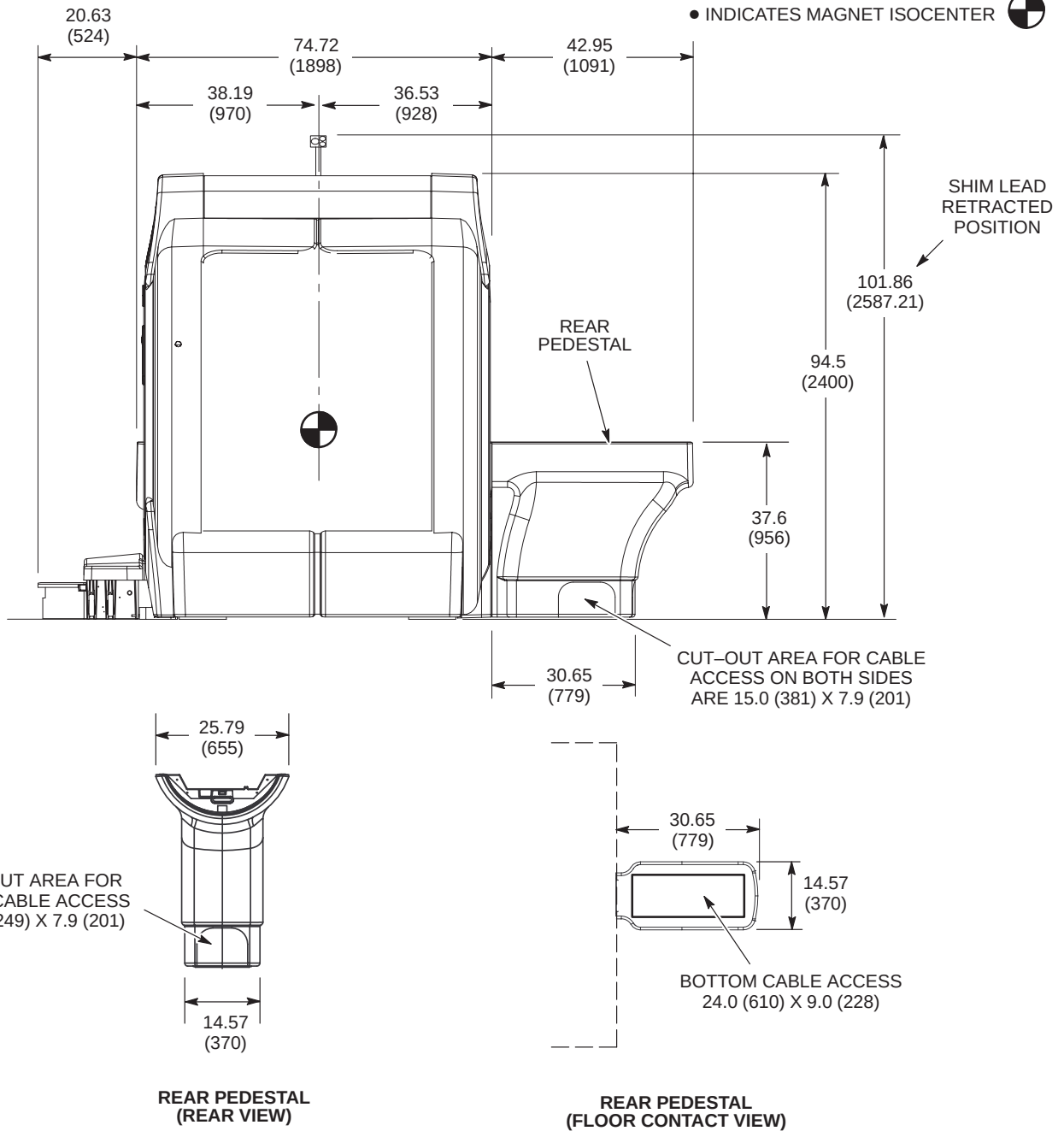
1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET WITH WIDOPEN ENCLOSURE (FRONT AND REAR VIEWS)

ILLUSTRATION 2-2

2-3 CABLING CONSIDERATIONS (Continued)

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- INDICATES MAGNET ISOCENTER



CABLE ACCESS FOR 1.0T AND 1.5T LCC (ACTIVE SHIELD) MAGNET WITH WIDOPEN ENCLOSURE
ILLUSTRATION 2-3

2-4 SPECIAL SITING CONSIDERATIONS

The following system components have special siting concerns which need to be considered.

2-4-1 Equipment For Gradient Coil Water Cooling

Note

All of the Gradient Coil water cooling solution (water/antifreeze mixture) must be replaced during the upgrade. The upgrade installs new equipment for Gradient Coil Cooling that requires new water/antifreeze mixture. Nitrogen gas must be supplied to be used to purge old cooling solution residue in the Gradient Coil, flexible vinly lines, and old heat exchanger before return shipment.

Fixed Site System

A dedicated, closed loop, chilled water cooling system for the Gradient Coil is provide with the upgrade for fixed site systems. The system consists of a water-to-air Water Chiller (WC1), two flexible vinyl hoses, and de-ionized water and anti-freeze (provided with the coil). This chiller replaces the Heat Exchanger (HE1) previously used for Gradient Coil cooling. See Illustration 2-4 for Water Chiller (WC1) dimension information. The two flexible hoses (supply and return) are to be routed from the Water Chiller (WC1) through waveguides in the Penetration Panel (PP1), through the Rear Pedestal (MG3), and connect to the rear of the Gradient Coil with supplied adjustable compression clamps. The Water Chiller will be powered from the PDU via a supplied power cord. Refer to Section 6, INTERCONNECT DATA, for additional cable information.

Note

The Fixed Site Water Chiller (WC1) for Gradient Coil Cooling must not be located below the floor level of the Gradient Coil. The Water Chiller reservoir tank has a low pressure vent plug which relieves pressure from thermal expansion of water and activates when tank pressure reaches 3-5 psi. The vent plug could allow fluid to leak if located below the MR system level. The reservoir tank vented plug can not be replaced with a non-vented type, damage to the tank may occur. The Water Chiller may be located one floor level above the MR system.

Mobile System (True Mobile, Transportable, Relocatable)

The Water Chiller (WC1) as shown in Illustration 2-4 is not used for Mobile configurations. The Mobile system utilizes a water-to-water heat exchanger for Gradient Coil Cooling which replaces the air-to-water Heat Exchanger (HE1) used in the Signa Horizon system. The Mobile van will provide chilled water for the new heat exchanger operation. The Van Manufactuer layout drawings will provided information for the water-to-water Heat Exchanger installation.

2-5 FLOOR LOADING AND WEIGHTS

This section contains loading considerations for upgrade components. Listed in Table 2-4 are the weights, floor loading, and normal mounting methods for components.

TABLE 2-4
SIGNA HORIZON FLOOR LOADING

COMPONENT	NET WT lbs (kg)	OVERALL DIMENSIONS W x D x H in. (mm)	LOAD PATTERN in. (mm)	NORMAL MOUNTING METHOD
LCC Magnet, Enclosure, and Coils	See Notes 1, 2 & 3	83.27 x 75.19 x 94.96 (2115 x 1910 x 2412)	Magnet load pattern is not changed but Enclosure & Rear Pedestal floor contact dimensions are different.	Magnet & Table Dock Assembly Resting on base, bolted to floor with RF Screen Room vendor supplied/installed bolts and anchors.
Water Chiller for Gradient Coil	374 (170)	21.65 x 27.95 x 37 (550 x 710 x 940)	Rectangular base 121.65 x 27.95 (550 x 710). Four casters 3 (76) dia. x .75 (19) wide.	Set on floor and rest on casters.
Note 1 For Signa MR/i: Weight of magnet with enclosure, RF/Gradient (BRM) Body Coil, and cryogenics is – 1.0T LCC (Active Shield) magnet 11,856 lbs (5378 kg) – 1.5T LCC (Active Shield) magnet 12,606 lbs (5719 kg) For Signa CV/i: Weight of magnet with enclosure, RF/Gradient (CRM) Body Coil, and cryogenics is – 1.5T LCC (Active Shield) magnet 12,806 lbs (5821 kg) 2 Consult a structural engineer on method of calculating proper weight/unit area for floor loading. 3 Refer to Section 2-2 MINIMUM DOOR/HALLWAY SIZES for Gradient Coil Assembly replacement weight and dimension requirements.				

2-6 COMPONENT DIMENSIONS

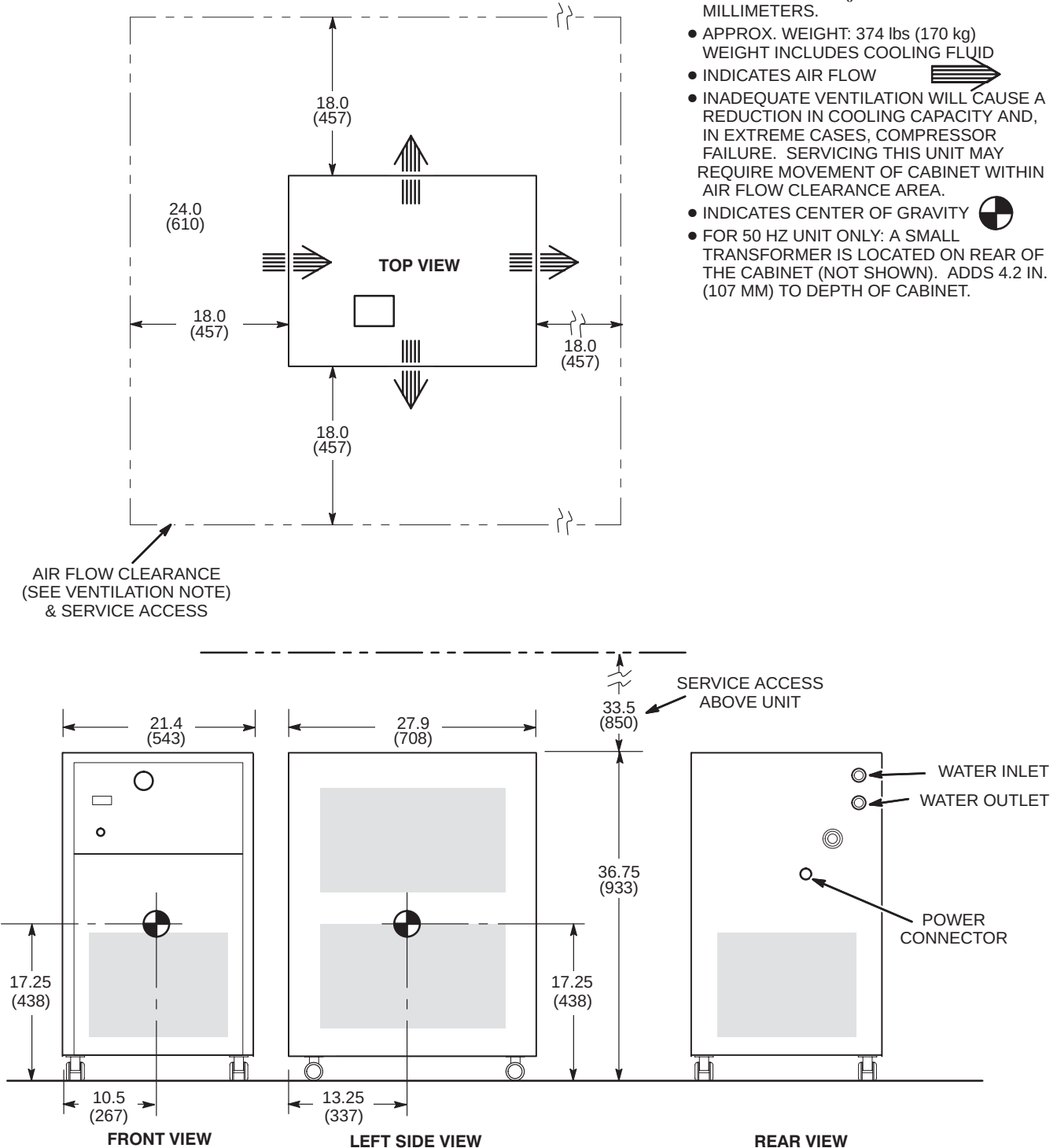
To assist in completing your room layout, refer to Table 2-5 for list of component Illustrations and the upgrade catalog numbers which provide the new equipment. Included are items which may impact site layout or need to be taken into consideration when determining delivery route. Modifications to existing cabinets are not included, more than one catalog may list the same new equipment but internally the equipment may be different due to other modification hardware.

TABLE 2-5
MR SYSTEM COMPONENT ILLUSTRATIONS LISTED BY UPGRADE CATALOG

UPGRADE CATALOG	ILLUSTRATION NAME	ILLUSTRATION NUMBER
M1085KB	WATER CHILLER (WC1) FOR GRADIENT COIL COOLING WATER	2-4
M1085KC	WATER CHILLER (WC1) FOR GRADIENT COIL COOLING WATER	2-4
M1090YL	STANDARD BRIDGE – LCC MAGNET WIDEOPEN ENCLOSURE	2-5
	FRONT ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA MR/i	2-7
	REAR ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA MR/i	2-8
	FRONT RIGHT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-11
	FRONT LEFT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-12
	REAR RIGHT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-13
	REAR LEFT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-14
	REAR PEDESTAL	See Note 1
M1090YN	STANDARD BRIDGE – LCC MAGNET WIDEOPEN ENCLOSURE	2-5
	FRONT ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA MR/i	2-7
	REAR ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA MR/i	2-8
	FRONT RIGHT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-11
	FRONT LEFT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-12
	REAR PEDESTAL	See Note 1
M1091YM	CRM BRIDGE – LCC MAGNET WIDEOPEN ENCLOSURE	2-6
	FRONT ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA CV/i	2-9
	REAR ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA CV/i	2-10
	FRONT RIGHT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-11
	FRONT LEFT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-12
	REAR RIGHT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-13
	REAR LEFT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE	2-14
	RF COIL CASE – LCC MAGNET WIDEOPEN ENCLOSURE USED FOR SIGNA MR/i & SIGNA CV/i COILS	2-15
	REAR PEDESTAL	See Note 1
M1085RF	RF COIL CASE – LCC MAGNET WIDEOPEN ENCLOSURE USED FOR 1.0T & 1.5T SIGNA MR/i UPGRADE	2-15
M1085RG	RF COIL CASE – LCC MAGNET WIDEOPEN ENCLOSURE USED FOR 1.5T SIGNA CV/i UPGRADE	2-16
M1885RF	RF COIL CASE – LCC MAGNET WIDEOPEN ENCLOSURE USED FOR 1.0T & 1.5T SIGNA MR/i UPGRADE	2-15
Note 1 Rear Pedestal is supplied with the indicated catalog numbers. Refer to Section 2-2, MINIMUM DOOR/HALLWAY SIZES, for dimension information.		

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 374 lbs (170 kg)
WEIGHT INCLUDES COOLING FLUID
- INDICATES AIR FLOW 
- INADEQUATE VENTILATION WILL CAUSE A REDUCTION IN COOLING CAPACITY AND, IN EXTREME CASES, COMPRESSOR FAILURE. SERVICING THIS UNIT MAY REQUIRE MOVEMENT OF CABINET WITHIN AIR FLOW CLEARANCE AREA.
- INDICATES CENTER OF GRAVITY 
- FOR 50 HZ UNIT ONLY: A SMALL TRANSFORMER IS LOCATED ON REAR OF THE CABINET (NOT SHOWN). ADDS 4.2 IN. (107 MM) TO DEPTH OF CABINET.

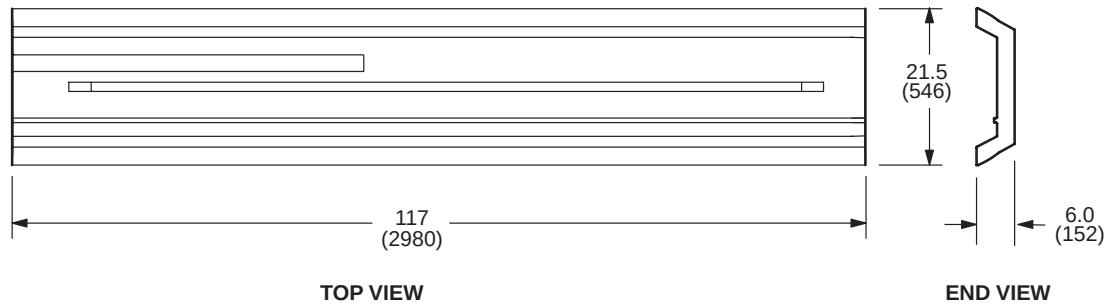


WATER CHILLER (WC1) FOR GRADIENT COIL COOLING WATER

ILLUSTRATION 2-4

NOTE:

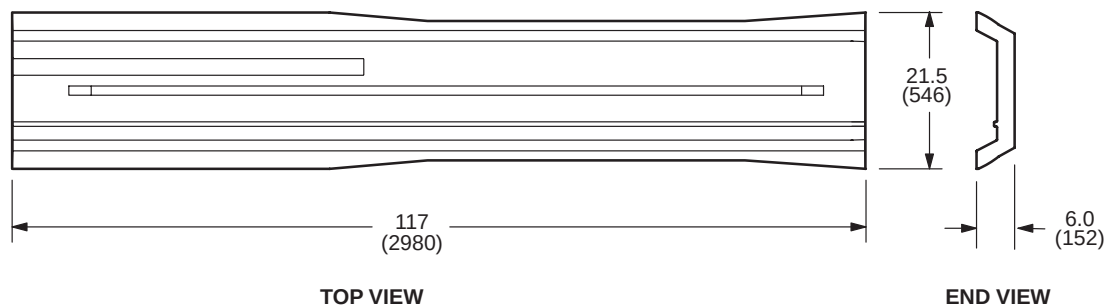
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



STANDARD BRIDGE – LCC MAGNET WIDEOPEN ENCLOSURE
ILLUSTRATION 2–5

NOTE:

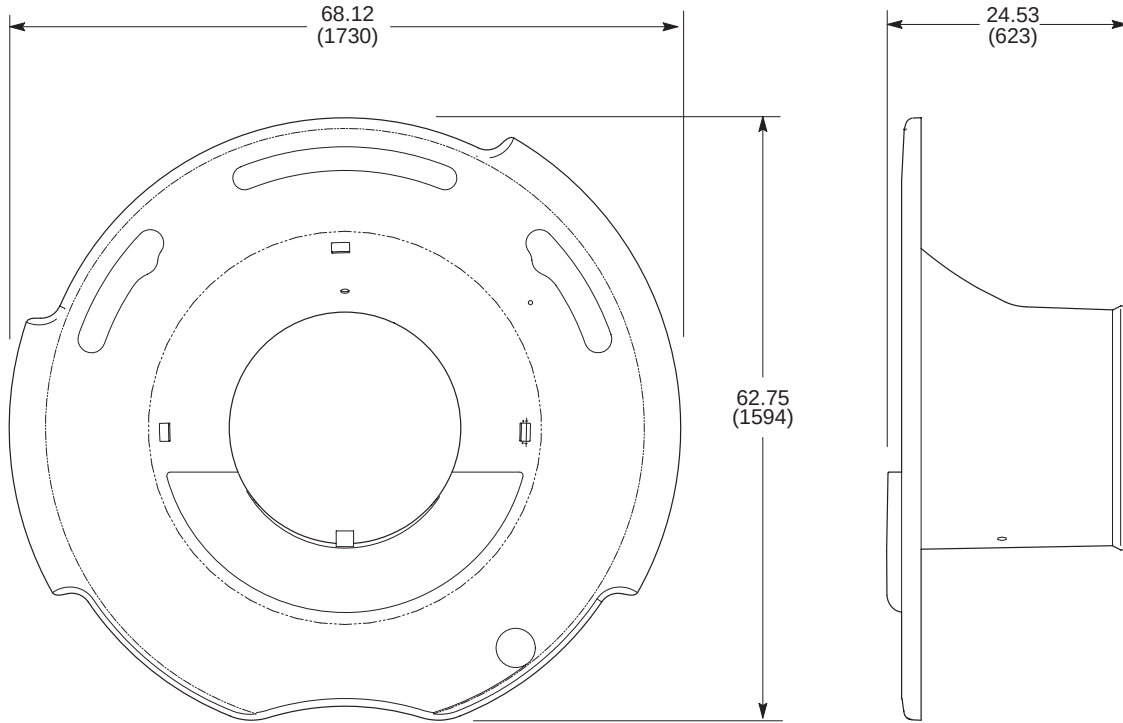
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



CRM BRIDGE – LCC MAGNET WIDEOPEN ENCLOSURE
ILLUSTRATION 2–6

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 70 lbs (32 kg)



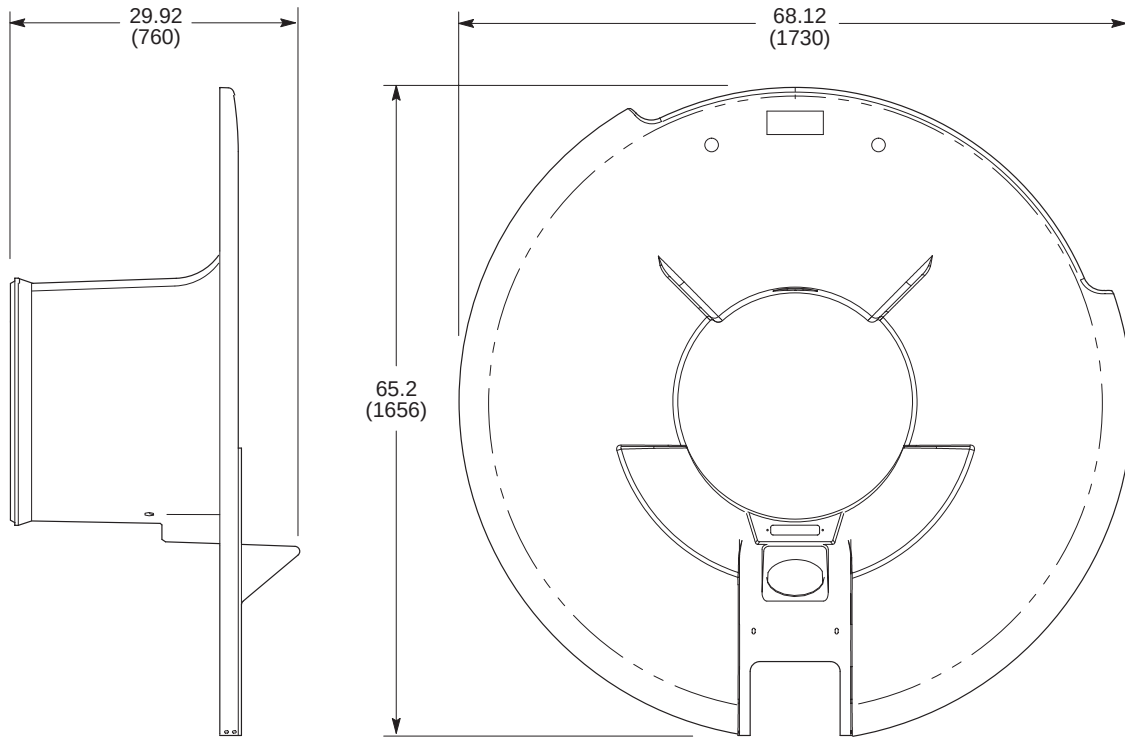
FRONT VIEW

SIDE VIEW

FRONT ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA MR/i
ILLUSTRATION 2-7

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



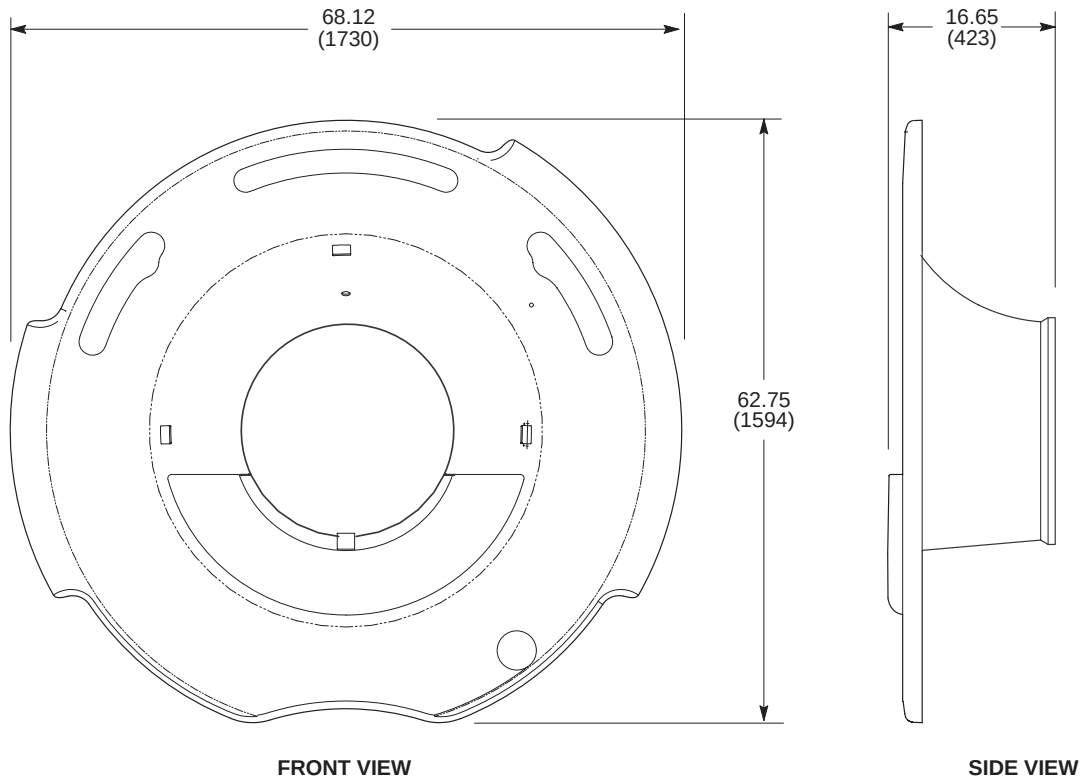
SIDE VIEW

FRONT VIEW

REAR ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA MR/i
ILLUSTRATION 2-8

NOTE:

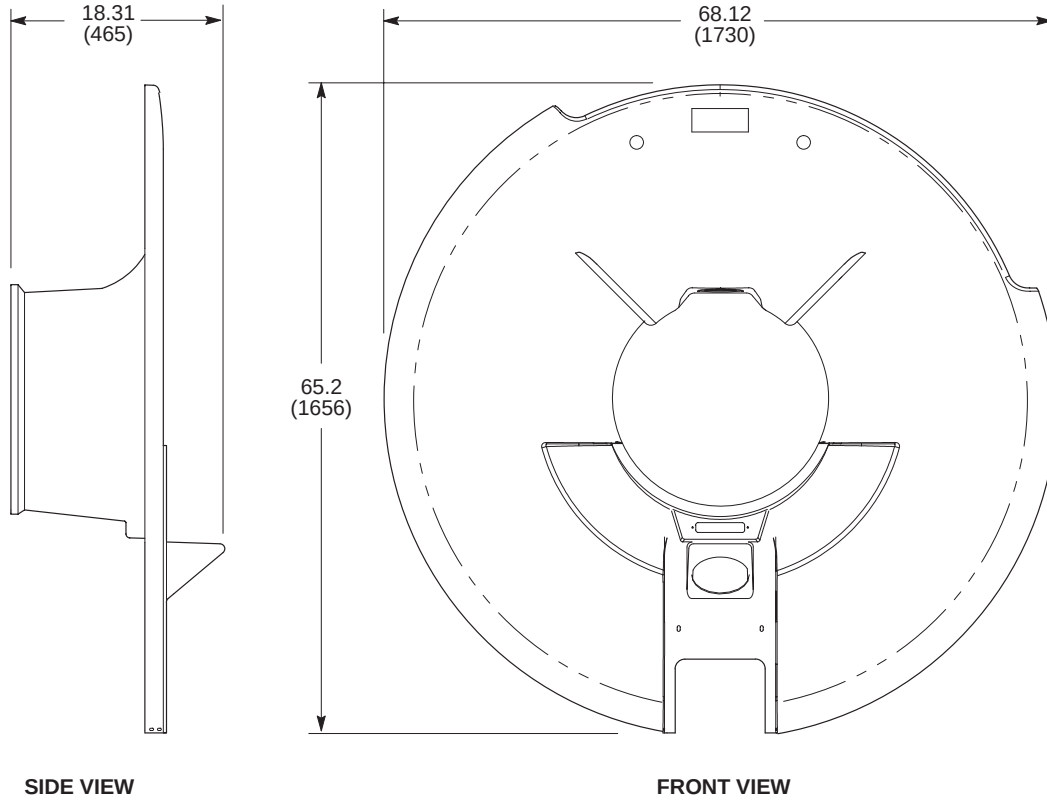
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 70 lbs (32 kg)



FRONT ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA CV/i
ILLUSTRATION 2–9

NOTE:

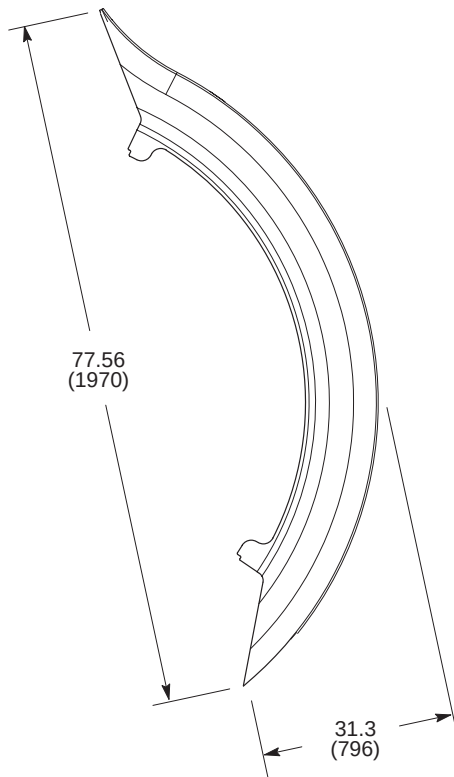
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 75 lbs (34 kg)



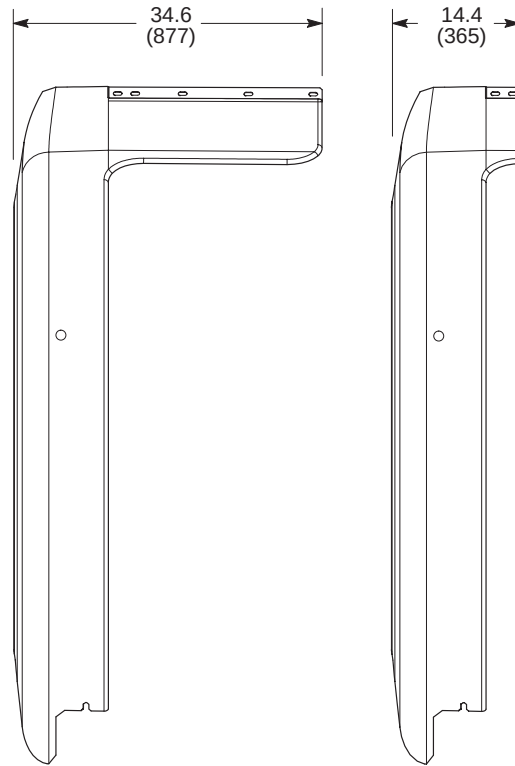
REAR ENDBELL – LCC MAGNET WIDEOPEN ENCLOSURE FOR SIGNA CV/i
ILLUSTRATION 2-10

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



FRONT VIEW



FIXED SITE

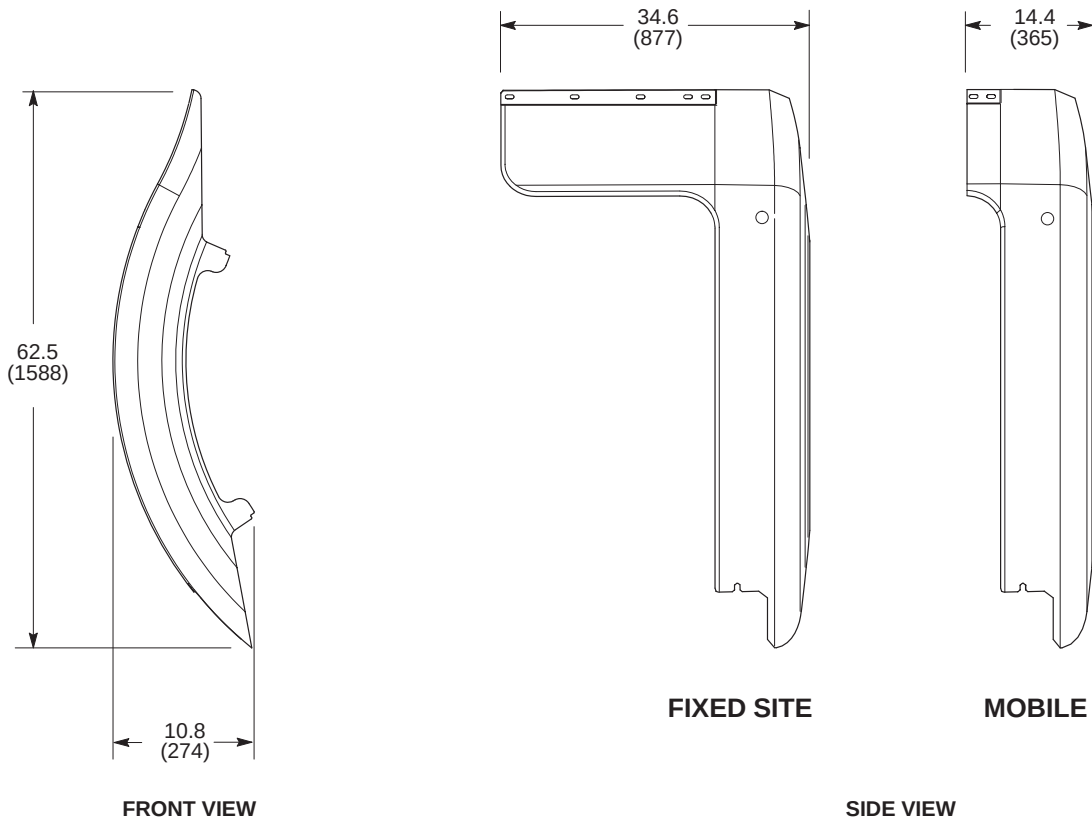
MOBILE

SIDE VIEW

FRONT RIGHT ARC COVER – LCC MAGNET WIDOPEN ENCLOSURE
ILLUSTRATION 2-11

NOTE:

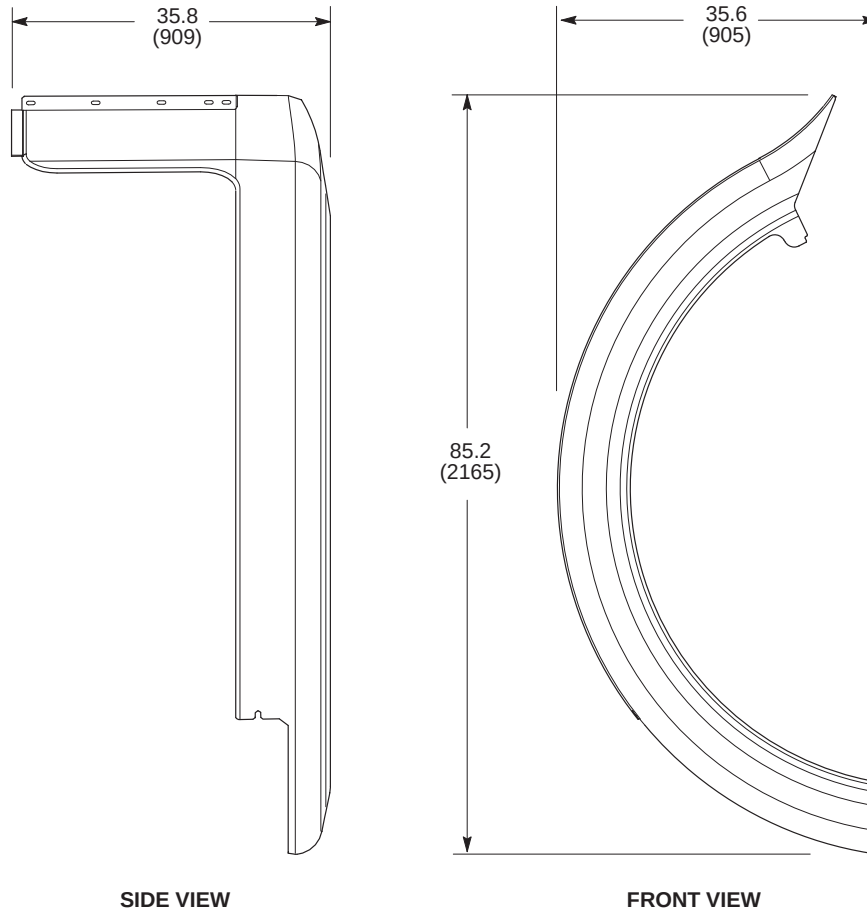
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



FRONT LEFT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE
ILLUSTRATION 2-12

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



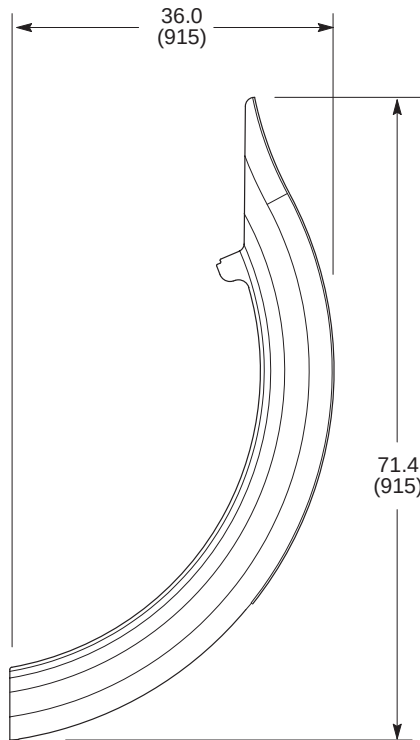
SIDE VIEW

FRONT VIEW

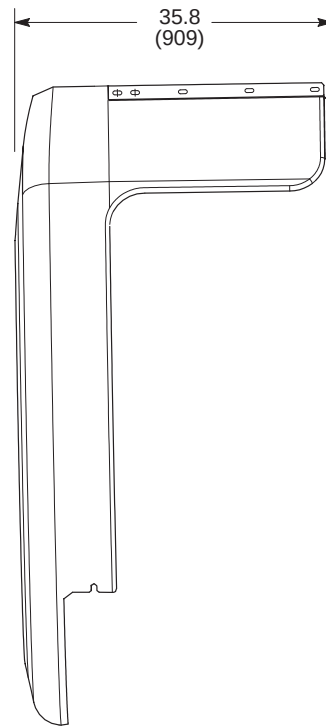
REAR RIGHT ARC COVER – LCC MAGNET WIDEOPEN ENCLOSURE
ILLUSTRATION 2-13

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 20 lbs (9 kg)



FRONT VIEW

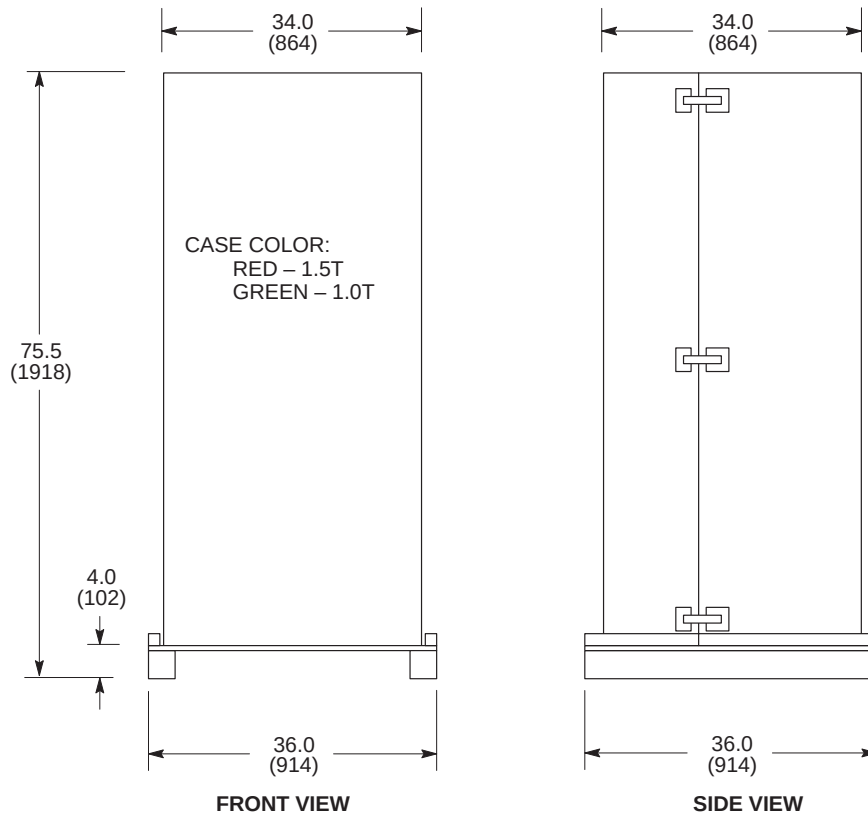


SIDE VIEW

REAR LEFT ARC COVER – LCC MAGNET WIDOPEN ENCLOSURE
ILLUSTRATION 2-14

NOTE:

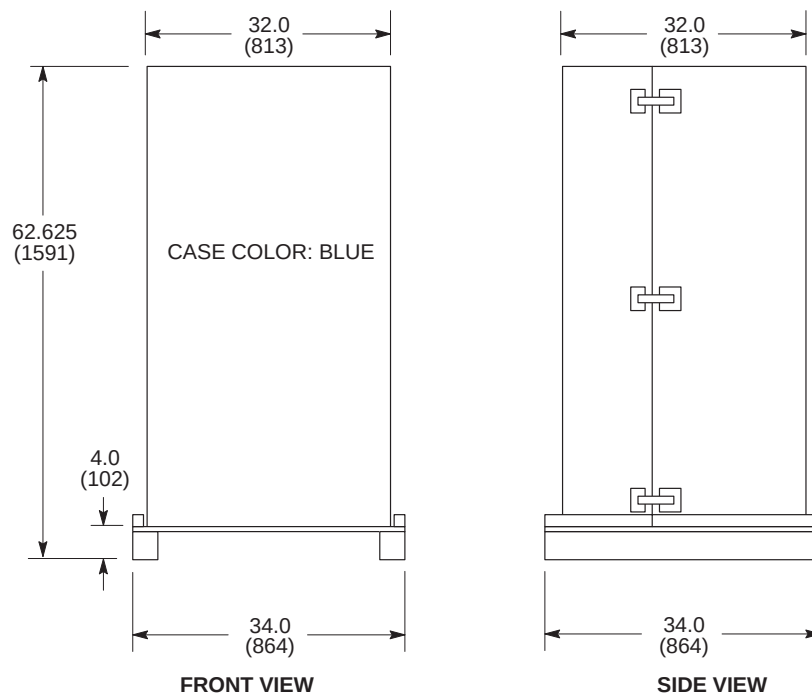
- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 275 lbs (125 kg)
WEIGHT AND DIMENSION INCLUDE COIL, CASE, AND SHPPING PALLET.
- RF COIL CASE HAS WHEELS AND HANDLES FOR EASE MOVING AFTER REMOVAL FROM PALLET.



RF COIL CASE – LCC MAGNET WIDOPEN ENCLOSURE USED FOR 1.0T & 1.5T SIGNA MR/i UPGRADE
ILLUSTRATION 2-15

NOTE:

- ALL DIMENSIONS ARE IN INCHES
ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.
- APPROX. WEIGHT: 275 lbs (125 kg)
WEIGHT AND DIMENSION INCLUDE COIL, CASE, AND SHPPING PALLET.
- RF COIL CASE HAS WHEELS AND HANDLES FOR EASE MOVING AFTER REMOVAL FROM PALLET.



RF COIL CASE – LCC MAGNET WIDEOPEN ENCLOSURE USED FOR 1.5T SIGNA CV/i UPGRADE
ILLUSTRATION 2-16